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Original Research

Deep learning-based prediction of acute pancreatitis severity from abdominal CT with multicenter external validation

Yanqi Xu, MSc[†,1], Brigitta Teutsch, PhD[†,2,3,4], Weicheng Zeng, MSc[1], Yang Hu, BS[5], Shikhar Rastogi, MSc[1], Emmy Yuebi Hu, MD[6], Isabella DeGregorio, MD[7], Wan Fung Chui, MD[6], Benjamin I. Richter, MD[8], Ryan Cummings, MD[6], Julia E. Goldberg, MD, MBA[6], Edwin Mathieu, MD[6], Belinda Appiah Asare, MD[6], Péter Hegedűs, MD[3], Kriszta-Beáta Gurza, MD[3], István Viktor Szabó, MD[3], Hedvig Tarján, MD[3], Andrea Szentesi, PhD[2,4], Ruben Borbély, MD[2,9], Dorottya Molnár, MD[9], Nándor Faluhelyi, MD, PhD[10], Áron Vincze, MD, PhD[11], Katalin Márta, MD, PhD[2,12], Péter Hegyi, MD, PhD[2,4,12,13], Qi Lei, PhD[1], Tamas Gonda, MD[14], Chenchan Huang, MD[†,6], Yiqiu Shen, PhD[*,†,1,6]

[1]Center for Data Science, New York University, New York, New York, 10011, United States
[2]Centre for Translational Medicine, Semmelweis University, Budapest, 1085, Hungary
[3]Department of Radiology, Medical Imaging Centre, Semmelweis University, Budapest, 1082, Hungary
[4]Institute for Translational Medicine, University of Pécs, Pécs, 7624, Hungary
[5]Gallatin School of Individualized Study, New York University, New York, New York , 10003, United States
[6]Department of Radiology, NYU Langone Health, New York, New York, 10016, United States
[7]Cooper Medical School of Rowan University, Camden, New Jersey, 08103, United States
[8]Division of Gastroenterology & Advanced Endoscopy, Department of Medicine, Rutgers New Jersey Medical SchoolNewark, New Jersey, 07103, United States
[9]Department of Medical Imaging, Bajcsy-Zsilinszky Hospital and Clinic, Budapest, 1105, Hungary
[10]Department of Medical Imaging, University of Pécs, Pécs, 7624, Hungary
[11]Division of Gastroenterology, First Department of Medicine, University of Pécs, Pécs, 7624, Hungary
[12]Institute of Pancreatic Diseases, Semmelweis University, Budapest, 1083, Hungary
[13]Translational Pancreatology Research Group, Interdisciplinary Centre of Excellence for Research Development and Innovation, University of Szeged, Szeged, 6720, Hungary
[14]Division of Gastroenterology and Hepatology, Department of Medicine, NYU Langone Health, New York, New York, 10016, United States
*Corresponding author: Yiqiu Shen, PhD, Department of Radiology, NYU Grossman School of Medicine, 227 East 30th Street, 745, New York, New York, 10016, United States (yiqiu.shen@nyulangone.org).
†These authors contributed equally as first authors. These authors contributed equally as senior authors.

Abstract

Background **Acute pancreatitis (AP) is a common gastrointestinal disease with a rising global incidence. While most cases are mild, severe AP (SAP) carries high mortality. Early and accurate severity prediction facilitates management optimization. However, existing clinical severity prediction models, such as Bedside Index of Severity in Acute Pancreatitis (BISAP) and Modified CT Severity Index (mCTSI), have modest accuracy and often rely on data unavailable at admission.**

Purpose **This study proposes a deep learning (DL) model to predict AP severity using abdominal contrast-enhanced CT scans acquired within 24 hours of admission.**

Materials and Methods **We collected 10 130 studies from 8335 patients across a multi-site U.S. health system and 3488 studies from public datasets. The DL model was trained in 2 stages: (1) self-supervised pretraining on 11 896 unlabeled studies and (2) fine-tuning on 550 labeled studies. Performance was evaluated against mCTSI on a hold-out internal test set (*$n = 100$* patients) and against both mCTSI and BISAP on an external Hungarian AP registry (*$n = 518$* patients).**

Results **On the internal test set, the model achieved areas under receiver operating curve (AUROCs) of 0.888 (95% CI: 0.800-0.960, sensitivity 0.50, specificity 0.92) for SAP and 0.888 (95% CI: 0.819-0.946, sensitivity 0.75, specificity 0.85) for mild AP (MAP), outperforming mCTSI (*$P = .002$*). External validation showed AUROCs of 0.887 (95% CI: 0.825-0.941, sensitivity 0.73, specificity 0.88) for SAP and 0.858 (95% CI: 0.826-0.888, sensitivity 0.38, specificity 0.98) for MAP, surpassing**

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mCTSI ($P = .024$) and BISAP ($P = .002$). In retrospective triage analysis, the model correctly identified 50%-73% of patients who progressed to SAP and 40%-73% of those with MAP.

Conclusion **The proposed DL model achieved performance comparable to or better than established prognostic tools and maintained robust external performance. These findings suggest that AI-assisted CT analysis may support early, automated risk stratification of AP.**

Keywords acute pancreatitis, deep learning, artificial intelligence, contrast-enhanced computed tomography, severity prediction

Abbreviations

AP = acute pancreatitis, RAC = revised atlanta classification, AI = artificial intelligence, DL = deep learning, CT = computed tomography, CECT = contrast-enhanced computed tomography, SAP = severe acute pancreatitis, MAP = mild acute pancreatitis, MSAP = moderately severe acute pancreatitis, BISAP = bedside index of severity in acute pancreatitis, mCTSI = modified computed tomography severity index, SSL = self-supervised learning, AUROC = area under receiver operating curve, AUPRC = area under the precision-recall curve, PPV = Positive Predictive Value, NPV = Negative Predictive Value

Summary

An AI model using admission-time CT scans accurately predicts acute pancreatitis severity long term during hospitalization, which may inform early clinical decision-making.

Key results

- Our AI model achieved strong generalization in an external international cohort (n = 518 patients), with AUROCs of 0.887 for predicting severe acute pancreatitis (SAP) and 0.858 for predicting mild acute pancreatitis (MAP), outperforming clinical scoring ($P < .05$).
- Self-supervised pretraining and pseudo-labeling improved performance for SAP prediction, increasing AUROC from 0.799 to 0.887 ($P < .001$).
- In retrospective triage, the model correctly identified 50%-73% of patients who progressed to SAP and 40%-73% of MAP cases.

Introduction

Acute pancreatitis (AP) is a common gastrointestinal disease characterized by sudden inflammation of the pancreas. In the United States, it ranks among the top causes of gastrointestinal-related hospitalizations, accounting for more than 275 000 hospital admissions annually.[1] Its global incidence continues to rise, imposing a growing burden on healthcare systems.[2] Although over 80% of cases are mild and self-limiting, ∼5% progress to severe AP (SAP), which carries a mortality rate as high as 30%-40%.[3]

The Revised Atlanta Classification (RAC) stratifies AP into mild (MAP), moderately severe AP (MSAP), and severe AP (SAP) based on the presence and duration of local or systemic complications.[4] However, these complications typically emerge 48-72 hours after symptom onset, limiting the opportunity for timely intervention.[5] Accurate prediction of disease severity at admission is essential to guide early management: enabling safe discharge of MAP patients, while preparing intensive care unit admission for those likely to progress to SAP,[5] especially in settings with limited resources.

Existing prognostic models are generally divided into 2 categories: multifactorial scoring systems based on clinical and laboratory data, and imaging-based models. Multifactorial scoring systems, such as the Bedside Index of Severity in Acute Pancreatitis (BISAP)[6] show moderate and inconsistent performance, with reported areas under receiver operating curve (AUROCs) ranging from 0.57 to 0.84[7,8] and require numerous input parameters. Imaging-based prognostic models, such as the Modified CT Severity Index (mCTSI),[9] are derived from radiologists' interpretation of CT findings (eg, necrosis extent, fluid collections), and suffer from high inter-observer variability.[10]

The majority of patients presenting to the emergency room with abdominal pain in the U.S. undergo abdominal CT imaging.[11] This presents an opportunity to leverage routinely acquired imaging for automated AP severity assessment. Recent studies have explored machine learning models to predict AP severity from CT imaging acquired in the early stage of AP, demonstrating improved accuracy over traditional scoring systems.[12–16] However, most were trained on single-center data without external validation, raising concerns about their generalizability.[12–15] Additionally, they often rely on manually annotated severity labels, which are labor-intensive and difficult to scale.[12–16]

This study aimed to develop and externally validate a DL model that predicts AP severity from admission contrast-enhanced CT (CECT) scans. The model leverages large-scale unlabeled imaging data through self-supervised pretraining followed by supervised fine-tuning using a smaller labeled dataset. We hypothesized that combining self-supervised learning (SSL)

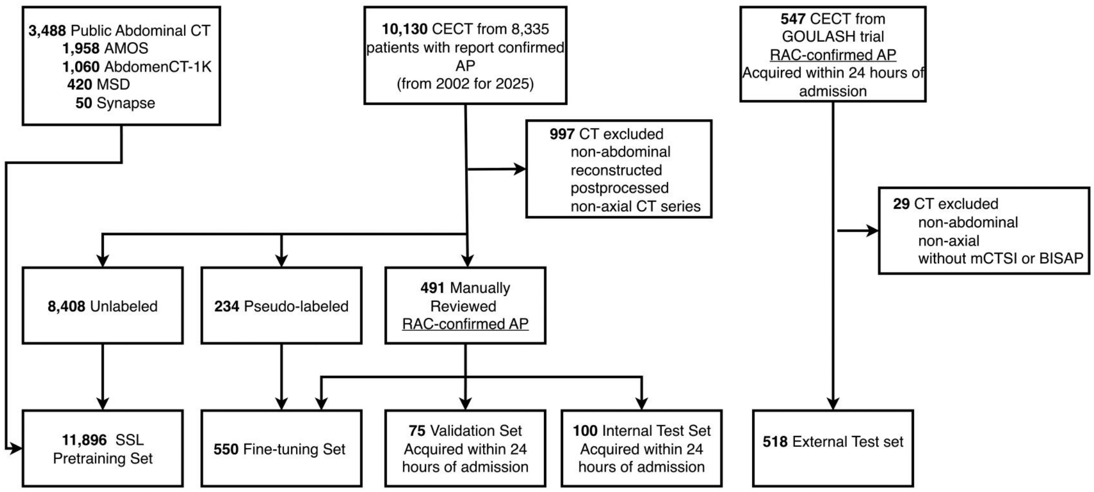

Figure 1 Flowchart of data cleaning. Non-abdominal CT series, reconstructed/post-processed images, and non-axial views were excluded from both internal and external cohorts. In total, 491 CECT studies in the internal cohort and all 547 studies in the external cohorts were manually reviewed, with AP diagnosis confirmed per RAC criteria. The DL model was pre-trained using SSL on 8408 unlabeled CT studies from the internal cohort and 3488 studies from 4 public abdominal CT datasets. Fine-tuning used 550 studies: 234 manually labeled (RAC criteria) and 316 pseudo-labeled from radiology reports (details in Supplement S2). Evaluation was performed on an internal test set of 100 studies and an external test set of 518 studies, both acquired within 24 hours of admission and manually annotated using RAC criteria.

and supervised learning would enhance predictive accuracy and generalizability by enabling the model to learn from larger and more diverse unlabeled data.

Materials and methods

This retrospective study was approved by NYU Langone Health Institutional Review Board (IRB Protocol I24-00008), and the requirement for informed consent was waived. All procedures are adhered to the ethical principles of the Declaration of Helsinki.

Data collection

The data collection and selection process for the development cohort (training and validation), internal test set, and external test set is illustrated in Figure 1. Details of the image acquisition protocol are described in Supplement S1.1, and image processing procedures are detailed in Supplement S1.2.

Development and internal test sets

We collected 10 130 CECT studies from 8335 patients between November 2002 and January 2025 at NYU Langone Health, a multi-hospital academic system that includes major tertiary centers and affiliated suburban hospitals. Eligibility criteria included age ≥18 years and a radiology report consistent with AP, fulfilling the imaging component of the RAC.[4]

To establish ground-truth labels, we curated a dataset of 491 cases using a stratified approach: 200 cases were randomly selected from the study population, while the remaining 291 were identified via targeted enrichment for clinical severity markers (ICU admission or length of stay >7 days). This strategy ensured adequate representation of SAP for model training and validation. Chart review confirmed that all 491 patients met the standard diagnostic criteria for AP, defined by abdominal pain or serum lipase ≥3× the upper limit of normal.[4] AP severity was retrospectively labeled as MAP, MSAP, and SAP, according to the RAC,[4] based on both CT imaging and EHR review. Modified CT Severity Index scores were independently assigned by 4 abdominal imaging fellows through direct review of CT images, blinded to clinical information, under the supervision of an abdominal radiologist (9 years of post-abdominal imaging fellowship experience). Details of the severity labeling and mCTSI scoring are provided in Supplement S1.3.

The 491 studies were randomly split into 316 for fine-tuning, 75 for validation, and 100 for internal testing. The split is performed at the patient level to prevent data leakage. To expand the fine-tuning set, 234 additional AP severity pseudo-labels were extracted from radiology reports using rule-based string-matching. The detailed procedures and validation of this approach are described in Supplement S2.

To enable self-supervised pretraining, we constructed an SSL dataset of 11 896 unlabeled CT studies, comprising the remaining 8408 studies from the internal dataset and 4 public abdominal CT datasets: AMOS[17] ($n = 1958$), AbdomenCT-1K[18] ($n = 1060$), MSD[19] ($n = 420$), and Synapse[20] ($n = 50$).

In total, the development cohort included 12 521 CT studies—11 896 for SSL, 550 for fine-tuning (316 manually labeled and 234 pseudo-labeled), and 75 for validation. Pseudo-labeled studies were used exclusively for fine-tuning and were not included in validation or testing. The 100 manually labeled studies in the internal test set were entirely independent of the development cohort. All validation and test CTs were acquired within 24 hours of hospital admission.

External test dataset

The external test dataset was obtained from the GOULASH trial, a multicenter randomized controlled study conducted at 3 tertiary medical centers in Hungary between 2017 and 2023.[21]

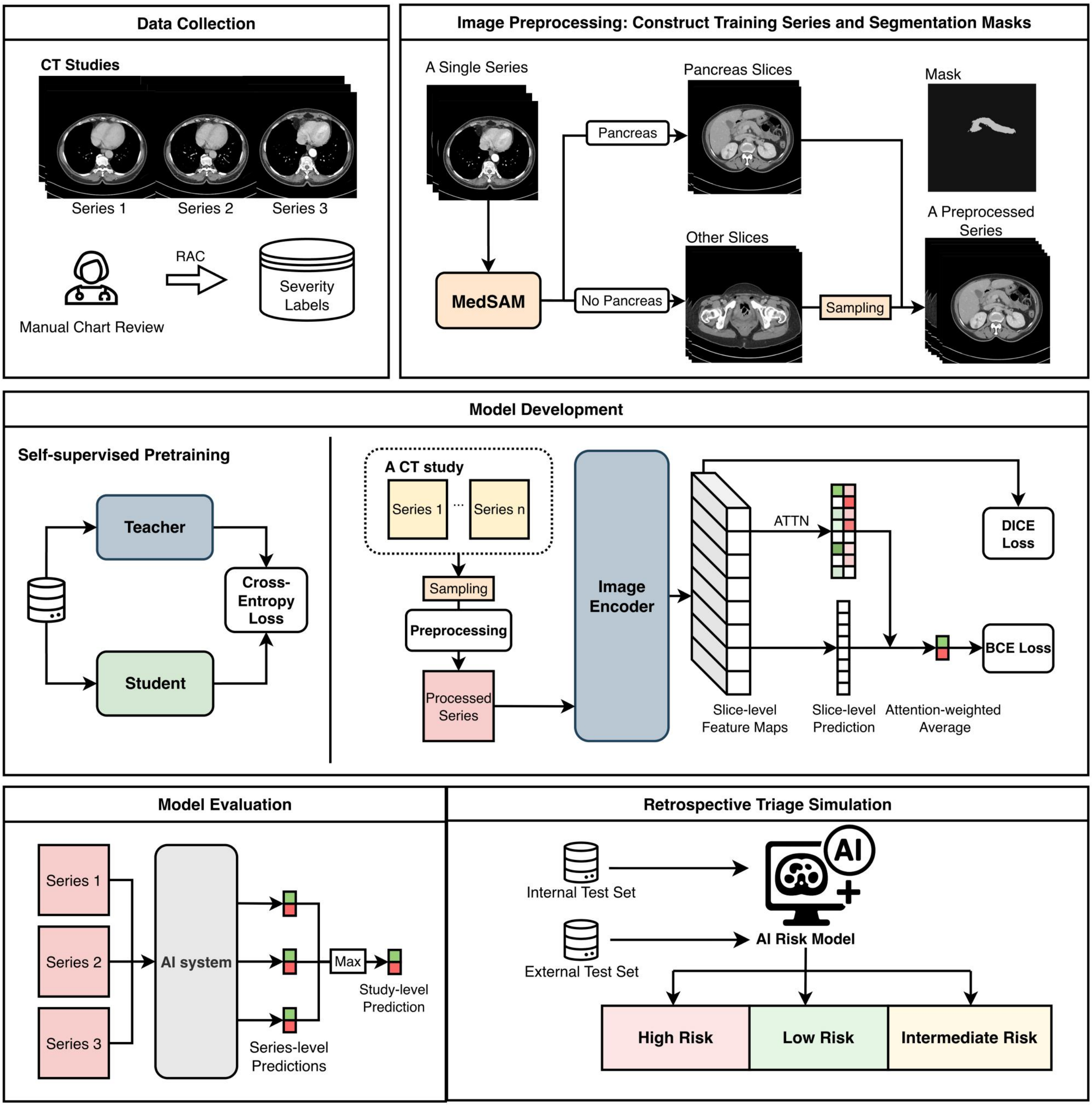

Figure 2 Workflow of the AI system. Data Collection: CT studies with multiple series were used in this study. Severity labels were assigned based on the RAC. Image Preprocessing: During training, MedSAM, an automated DL segmentation model, was used to generate pancreas masks and identify relevant slices. These were combined with randomly sampled non-pancreas slices to create preprocessed input series. Segmentation is not required at inference time. Model Development: A transformer-based backbone (pretrained via SSL) extracted slice-level features. Slice scores derived from saliency maps were aggregated using attention (ATTN) to generate severity predictions, optimized with binary cross entropy, Dice, and L1 losses. Model Evaluation: Study-level predictions were generated by using the maximum series-level score. A retrospective triage simulation was performed on both internal and external test sets. More details about image preprocessing and model development can be found in Supplement S3.

Inclusion criteria were: age over 18, AP diagnosis based on the RAC (diagnosed prospectively by medical doctors), no history of pancreatic malignancy, and availability of CT imaging within 24 hours of admission. Among 619 enrolled patients, 547 abdominal CT studies were available. After excluding non-abdominal series, non-axial images, and studies without valid mCTSI or BISAP scores, 518 CT studies remained. AP severity was retrospectively determined at discharge by a multidisciplinary team, including gastroenterology residents and specialists, following RAC criteria. The mCTSI score was retrospectively assessed by 6 radiology residents under attending radiologist supervision, all of whom were blinded to the clinical information.

Table 1 Statistics of the development cohort and external cohort.

	Development and Internal Test Sets				External Test Set
	Fine-tuning	Validation	Test	Total	Total
Patients	528	75	100	703	518
CT studies	550	75	100	725	518
Severity					
MAP, no. (%)	245 (44.6)	31 (41.3)	40 (40.0)	316 (43.6)	276 (53.3)
MSAP, no. (%)	183 (33.3)	37 (49.3)	48 (48.0)	268 (37.0)	209 (40.4)
SAP, no. (%)	122 (22.2)	7 (9.3)	12 (12.0)	141 (19.5)	33 (6.4)
Demographics					
Gender, female, no. (%)	222 (42.0)	32 (42.7)	41 (41.0)	295 (42.0)	212 (40.9)
Age, mean (SD)	52.88 (18.2)	51.53 (16.9)	54.33 (18.0)	52.94 (18.0)	54.42 (14.8)
Height, mean (SD), cm	1.69 (0.1)	1.68 (0.1)	1.67 (0.1)	1.69 (0.1)	1.70 (0.1)
Weight, mean (SD), kg	82.28 (21.2)	79.43 (19.0)	80.62 (18.0)	81.64 (20.4)	82.39 (17.1)
Imaging Findings					
Has pleural fluid, no. (%)	32 (13.8)	9 (12.0)	15 (15.0)	56 (13.8)	37 (7.1)
Has peripancreatic fluid collection, no. (%)	107 (46.1)	38 (50.7)	36 (36.0)	181 (44.5)	358 (69.1)
Has abdominal fluid, no. (%)	102 (44.1)	27 (36.0)	45 (45.0)	174 (42.8)	244 (47.1)

MAP, mild acute pancreatitis; MSAP, moderately severe acute pancreatitis; SAP, severe acute pancreatitis.

AI model

The workflow for AI model training and evaluation is illustrated in Figure 2. We used an off-the-shelf segmentation model, medSAM,[22] to automate the process of identifying pancreas-containing slices (Supplement S3.1). To improve generalizability, we first applied the DINO framework[23,24] to pretrain a Vision Transformer (ViT)[25] on the SSL dataset (Supplement S3.4). The pretrained ViT was then fine-tuned using the fine-tuning dataset. The model produces 2 outputs: the exam-level probabilities of MAP and SAP. Model architecture and training details are provided in Supplement S3. To leverage the complementary strengths of AI and established prognostic tools, we developed a hybrid prediction strategy integrating AI probabilities with mCTSI and BISAP scores. Detailed methodology is provided in Supplement S6.

Retrospective simulation

We conducted a retrospective simulation to stratify patients into 3 risk categories based on AI-predicted probabilities:

- High risk: patients with a probability of SAP above threshold ($\widehat{y}_{SAP} > t_S$) were classified as high-risk and considered for ICU admission.
- Low risk: patients with a probability of SAP below or equal to the threshold ($\widehat{y}_{SAP} <= t_S$) and a probability of MAP above threshold ($\widehat{y}_{MAP} > t_M$) were classified as low-risk, suggesting eligibility for conservative management (standard inpatient supportive care without organ failure support or invasive interventions).[26]
- Intermediate risk: remaining patients were labeled intermediate-risk, warranting standard inpatient monitoring and evaluation.

Thresholds, t_M for low-risk and t_S for high-risk, were selected from the internal validation set to jointly optimize sensitivity and positive predictive value (PPV), with the goal of surpassing mCTSI performance.

Qualitative case study

Gradient-weighted Class Activation Mapping[27] was used to generate heatmaps highlighting regions that contributed to the model's predictions. A board-certified abdominal radiologist with 9 years of experience reviewed saliency maps of 20 randomly selected SAP cases in the external test set.

Model evaluation and statistical analysis

Model performance was evaluated using AUROC, area under the precision-recall curve (AUPRC), sensitivity, specificity, PPV, and negative predictive value (NPV), with 95% CIs computed via 10 000 bootstrap samples. The operating point selection process is described in Supplement S4. Results are based on an ensemble of the top 5 models with the highest AUROC on the validation set of the development cohort. AUROC comparisons between the AI model, mCTSI, and BISAP were conducted using the DeLong test[28]; subgroup comparisons (Supplement S5) used unpaired DeLong tests. Sensitivity and specificity were compared using the permutation test. Formal correction for multiple comparisons was not applied, since the primary statistical comparisons in this study were prespecified and hypothesis-driven rather than exploratory. All metrics were computed using the scikit-learn package (v1.6.1), and DeLong tests were executed using the pROC package[29] (v1.18.5) in R (v4.5.1).

Publicly available datasets, including AMOS, AbdomenCT-1K, MSD, and Synapse, can be accessed from their respective repositories. The model weights and internal dataset are proprietary and cannot be shared publicly. Requests for access to the Goulash trial data should be directed to the study organizers. The model's inference code is released on GitHub under the MIT license at https://github.com/nyu-shenlab/ap-severity-evaluation.

Table 2 Performance Comparison of the AI model, the mCTSI, and the hybrid model on the internal test set.

		AUROC	AUPRC	Sensitivity	Specificity	PPV	NPV
SAP vs non-SAP	mCTSI	0.704 [0.551, 0.838]	0.205 [0.100, 0.414]	0.25 [0, 0.5]	0.920 [0.859, 0.976]	0.3 [0, 0.600]	0.9 [0.833, 0.956]
	AI	0.888 [0.800, 0.960]	0.566 [0.305, 0.831]	0.500 [0.200, 0.800]	0.920 [0.859, 0.976]	0.462 [0.182, 0.750]	0.931 [0.874, 0.978]
	Hybrid (AI + mCTSI)	0.891 [0.809, 0.973]	0.578 [0.308, 0.840]	0.500 [0.200, 0.786]	0.955 [0.907, 0.989]	0.600 [0.273, 0.909]	0.933 [0.878, 0.978]
MAP vs non-MAP	mCTSI	0.816 [0.733, 0.890]	0.688 [0.554, 0.812]	0.675 [0.525, 0.818]	0.850 [0.755, 0.933]	0.750 [0.600, 0.885]	0,797 [0.694, 0.892]
	AI	0.888 [0.819, 0.946]	0.805 [0.672, 0.928]	0.750 [0.610, 0.878]	0.850 [0.754, 0.933]	0.769 [0.629, 0.896]	0.836 [0.737, 0.924]
	Hybrid (AI + mCTSI)	0.897 [0.834, 0.960]	0.810 [0.673, 0.933]	0.875 [0.763, 0.973]	0.850 [0754, 0.934]	0.800 [0.667, 0.909]	0.911 [0.827, 0.981]

Evaluation metrics include AUROC, AUPRC, sensitivity, specificity, PPV, and NPV for identifying MAP and SAP. 95% CI reported within brackets.

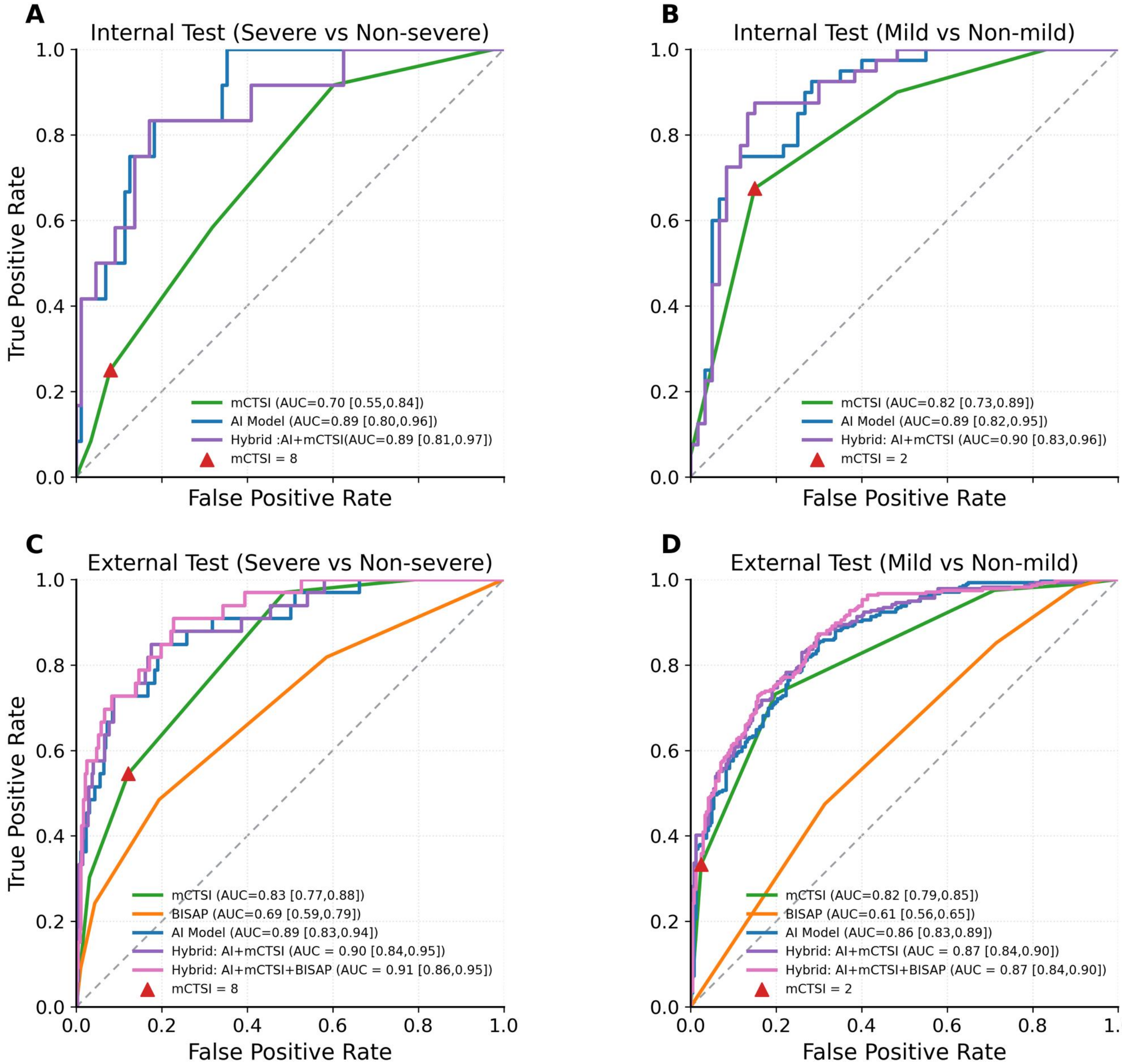

Figure 3 ROC Curves for the AI Model, mCTSI, and BISAP on (A & B) Internal ($n=100$ patients) and (C & D) External ($n=518$ patients) test sets. Operating points used for mCTSI risk stratification are indicated on the curves. On both datasets and across both tasks, the AI model outperformed mCTSI. On the external test set, it also outperformed BISAP. AUC's are reported with 95% CI within brackets.

Results

Table 1 summarizes the datasets used in this study. The internal test set included 100 abdominal CT studies from 100 patients (41% female; mean age: 54.33 ± 18.0 years), with severity labels distributed as follows: 40 MAP (40%), 48 MSAP (48%), and 12 SAP (12%). The external test set comprised 518 abdominal CT studies from 518 patients (40.9% female; mean age: 54.42 ± 14.8 years), with 276 MAP (53.3%), 209 MSAP (40.4%), and 33 SAP (6.4%).

Performance on the internal test set

For identifying high-risk patients (SAP v.s. non-SAP), the AI model achieved an AUROC of 0.888 (95% CI: 0.800-0.960), outperforming mCTSI (AUROC 0.704; 95% CI: 0.556-0.839; $P=.002$). The mCTSI showed specificity of 0.92 (95% CI: 0.859-0.976) and a sensitivity of 0.25 (95% CI: 0.0-0.5). At the same specificity, the AI model achieved a sensitivity of 0.50 (95% CI: 0.200-0.800, $P=.123$). For identifying low-risk patients (MAP vs non-MAP), the AI model achieved an AUROC of 0.888 (95% CI: 0.819-0.946), outperforming mCTSI (AUROC: 0.816, 95% CI: 0.733-0.890; $P<.001$). At the matched specificity (0.850; 95% CI: 0.755-0.933), the AI model achieved a sensitivity of 0.75 (95% CI: 0.610-0.878) compared to mCTSI (0.675, 95% CI: 0.525-0.818, $P=.291$). The hybrid AI-mCTSI model achieved AUROCs of 0.891 (95% CI: 0.809-0.973) for SAP and 0.897 (95% CI: 0.834-0.960) for MAP. The AI model showed no statistically significant performance differences across patient cohorts stratified by sex or age

Table 3 Performance comparison of the AI model, the BISAP, the mCTSI, and the hybrid model on the external test set.

		AUROC	AUPRC	Sensitivity	Specificity	PPV	NPV
SAP vs non-SAP	mCTSI	0.828 [0.769, 0.883]	0.231 [0.135, 0.363]	0.545 [0.375, 0.718]	0.878 [0.849, 0.907]	0.234 [0.141, 0.333]	0.966 [0.948, 0.982]
	BISAP	0.693 [0.592, 0.787]	0.152 [0.082, 0.272]	0.242 [0.103, 0.4]	0.957 [0.938, 0.974]	0.276 [0.120, 0.444]	0.949 [0.929, 0.967]
	AI	0.887 [0.825, 0.941]	0.442 [0.290, 0.639]	0.727 [0.567, 0.875]	0.878 [0.848, 0.907]	0.289 [0.195, 0.388]	0.979 [0.965, 0.991]
	Hybrid (AI + mCTSI)	0.895 [0.835, 0.947]	0.489 [0.335, 0.685]	0.727 [0.568, 0.875]	0.876 [0.846, 0.905]	0.286 [0.192, 0.385]	0.979 [0.965, 0.991]
	Hybrid (AI + mCTSI +BISAP)	0.913 [0.864, 0.954]	0.521 [0.352, 0.699]	0.727 [0.567, 0.875]	0.913 [0.888, 0.938]	0.383 [0.250, 0.483]	0.980 [0.966, 0.991]
MAP vs non-MAP	mCTSI	0.819 [0.785, 0.851]	0.797 [0.752, 0.837]	0.333 [0.279, 0.390]	0.975 [0.954, 0.992]	0.939 [0.888, 0.981]	0.562 [0.515, 0.609]
	BISAP	0.609 [0.564, 0.654]	0.560 [0.548, 0.652]	0.851 [0.809, 0.891]	0.285 [0.229, 0.343]	0.576 [0.529, 0.623]	0.627 [0.535, 0.713]
	AI	0.858 [0.826, 0.888]	0.866 [0.824, 0.905]	0.380 [0.324, 0.439]	0.975 [0.953, 0.992]	0.946 [0.899, 0.983]	0.580 [0.532, 0.628]
	Hybrid (AI + mCTSI)	0.874 [0.844, 0.902]	0.886 [0.849, 0.919]	0.402 [0.334, 0.460]	0.975 [0.953, 0.992]	0.949 [0.905, 0.984]	0.588 [0.540, 0.636]
	Hybrid (AI + mCTSI +BISAP)	0.873 [0.842, 0.902]	0.883 [0.844, 0.916]	0.337 [0.282, 0.393]	0.975 [0.953, 0.992]	0.939 [0.888, 0.980]	0.563 [0.514, 0.610]

Evaluation metrics include AUROC, AUPRC, sensitivity, specificity, PPV, and NPV for identifying MAP and SAP. 95% CI reported within brackets.

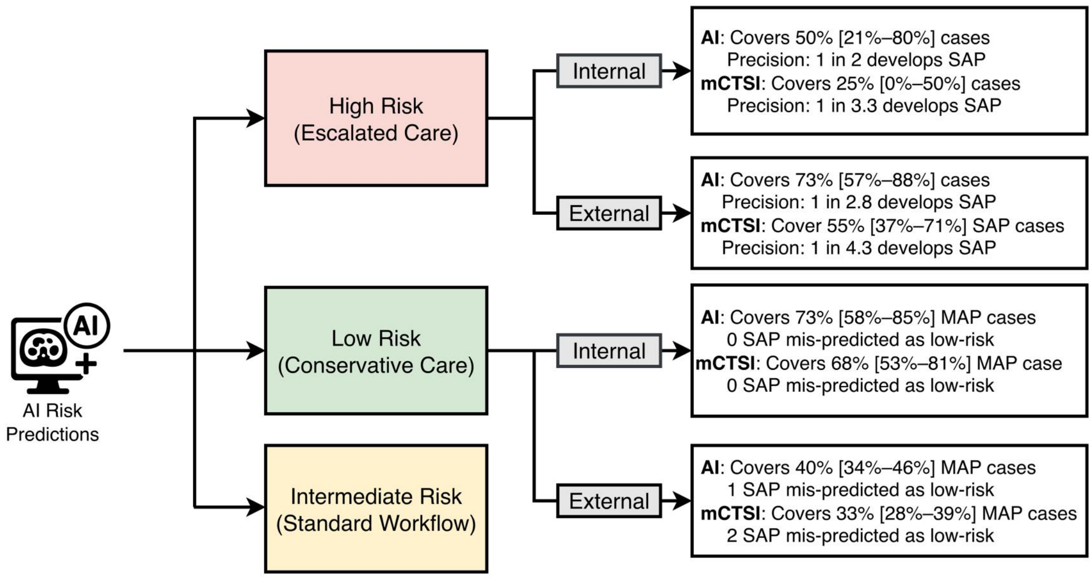

Figure 4 Clinical Simulation of Risk Stratification Using AI and mCTSI predictions. Patients were categorized into low-risk, intermediate-risk, and high-risk groups based on risk scores from the AI model and mCTSI. On the internal test set ($n = 100$), the AI model correctly identified 73% of MAP patients as low risk without misclassifying any severe cases. On the external test set ($n = 518$), the AI model identified 73% of SAP patients as high risk, with a precision of 1 severe case per 2.8 high-risk cases. 95% CI are in brackets. These results highlight the AI model's potential to improve risk stratification and support clinical decision-making.

(details in Supplement S5 and Table S3). Model performance on the internal test set is summarized in Table 2 and Figure 3.

Performance on the external test set

The AI model maintained consistent performance on the external test set, with no significant degradation compared with internal evaluation. For identifying high-risk cases (SAP vs non-SAP), the AI model achieved an AUROC of 0.887 (95% CI: 0.825-0.941), outperforming mCTSI (0.828, 95% CI: 0.769-0.883; $P = .024$) and BISAP (0.693, 95% CI: 0.592-0.787; $P = .002$). Similarly, for identifying low-risk patients (MAP vs non-MAP), the model achieved an AUROC of 0.858 (95% CI: 0.826-0.888), surpassing mCTSI (0.819; 95% CI: 0.785-0.851; $P = .019$) and BISAP (0.609; 95% CI: 0.564-0.654; $P < .001$). The hybrid model (AI + mCTSI + BISAP) achieved AUROCs of 0.913 (95% CI: 0.864-0.954, vs AI standalone $P = .11$) for SAP and 0.873 (95% CI: 0.842-0.902, vs AI standalone $P = .03$) for MAP. Model performance on the external test set is summarized in Figure 3, with additional metrics provided in Table 3.

Impact of SSL and pseudo labels

Ablation studies on the external test set evaluated the effects of SSL pretraining and pseudo-labeled data on model performance. The baseline model trained without SSL or pseudo-labeled data achieved an AUROC of 0.799 (95% CI, 0.717-0.871) for identifying SAP. Incorporating pseudo-labeled data increased AUROC to 0.845 (95% CI, 0.770-0.907; P = .007), and using SSL-pretrained weights increased it to 0.846 (95% CI, 0.773-0.906; $P < .001$). Combining both approaches achieved the highest AUROC of 0.887 (95% CI, 0.825-0.941; $P < .001$). Detailed metrics and results on MAP are provided in Table S4.

Risk-based triage simulation

The performance of retrospective analysis is shown in Figure 4. In the internal test set, the AI model classified 12% (95% CI, 6%-19%) of all patients as high risk and 34% (95% CI, 25%-43%) as low risk. It identified 50% (95% CI, 21%-80%) of SAP cases as high risk, compared with 25% (95% CI, 0%-50%; P = .12) by mCTSI. Among patients classified as high risk, the PPV was 0.50 (95% CI, 0.20-0.80) for the AI model and 0.30 (95% CI, 0.00-0.63; P = .11) for mCTSI. For low-risk triage, the AI model correctly identified 73% (95% CI, 58%-85%) of MAP cases, compared with 68% (95% CI, 53%-81%; P = .40) for mCTSI. Neither method misclassified SAP cases as low risk.

In the external test set, the AI model classified 13% (95% CI, 10%-15%) of patients as high risk and 23% (95% CI, 19%-27%) as low risk. It identified 73% (95% CI, 57%-88%) of SAP cases as high risk, compared with 55% (95% CI, 37%-71%; P = .04) for mCTSI. The AI model demonstrated higher precision, with one SAP case per 2.8 high-risk predictions (PPV, 0.36; 95% CI, 0.25-0.48), compared with one SAP case per 4.3 predictions for mCTSI (PPV, 0.23; 95% CI, 0.14-0.33; P = .002). For low-risk triage, the AI model correctly identified 40% (95% CI, 34%–46%) of MAP cases and misclassified one SAP case as low risk; mCTSI identified 33% (95% CI, 28%-39%; P = .02) of MAP cases and also misclassified one SAP case as low risk.

Qualitative case study

Qualitative analysis found that the model predominantly focused on imaging features clinically associated with SAP, including pancreatic necrosis, peripancreatic fluid or necrotic collections, fat stranding, and ascites. Figure 5 shows 3 representative cases with different severity grades. Additional true positives and false negatives are demonstrated in Figure S2. In

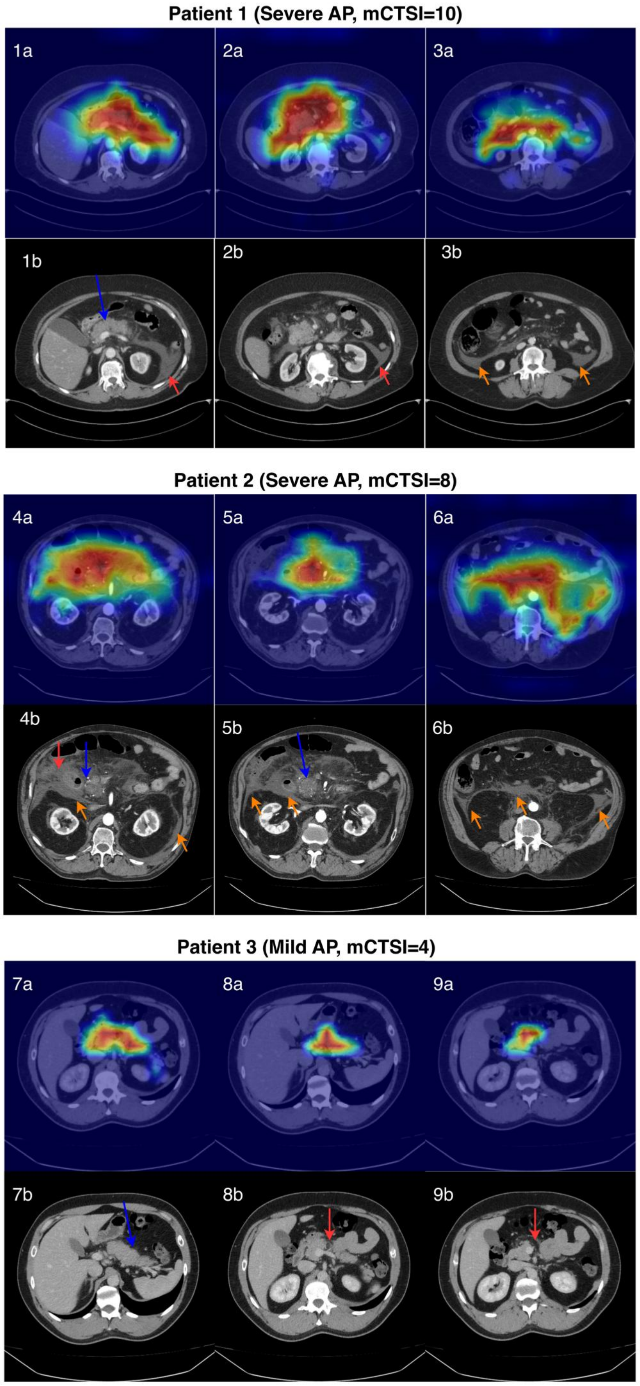

Figure 5 Grad-CAM heatmaps and corresponding CECT Images across AP severities. Top row (A): Grad-CAM heatmaps generated by the AI model. Bottom row (B): Corresponding axial portal-venous phase CT images. **Patient 1**: This case was correctly identified as severe by the AI model but was classified as moderately severe by mCTSI (score = 6). Each CT slice shows: (1) homogeneous enhancement of the pancreatic parenchyma (blue arrow), without pancreatic parenchymal necrosis (0 points); (2) acute peripancreatic fluid collection along the bilateral paracolic gutters (red arrows) (4 points), and (3) distant ascites more inferiorly in the abdomen (orange arrows) (2 points). **Patient 2**: Axial CECT image demonstrates: (1) hypoenhancement of the pancreatic head parenchyma (blue arrows), most compatible with pancreatic necrosis (2 points); (2) peripancreatic fat necrosis (red arrows) (4 points); and (3) distant ascites (orange arrows) (2 points), yielding a CTSI score of 8, consistent with SAP. **Patient 3**: Axial CECT image demonstrates diffuse pancreatic enlargement with loss of normal lobulation (blue arrow) and trace adjacent mesenteric fat stranding (red arrows), corresponding to a CTSI score of 2 and compatible with mild acute pancreatitis. The heatmaps indicate the model makes correct predictions, highlighting the pancreatic inflammation, acute peripancreatic fluid collection, and distant ascites, respectively, across different severities.

cases misclassified by both methods, neither the Grad-CAM heatmaps nor the CT images demonstrated clear radiologic features accounting for the severe clinical course.

Discussion

In this multicenter prognostic study, an AI model using admission CECT scans was developed to predict the severity of AP. The model incorporated SSL and pseudo-labeled data to leverage large volumes of unlabeled imaging studies, reducing dependence on manual annotation. Trained on more than 12 000 scans from a U.S. multi-hospital system and externally validated on a multicenter randomized trial from Hungary, the model maintained robust performance across diverse imaging protocols and patient populations. Compared with the mCTSI and the BISAP, it demonstrated higher discriminative ability for both mild and SAP.

Although clinical scoring systems such as BISAP facilitate early bedside assessment, our results suggest that imaging-based AI captures complementary prognostic signals beyond systems that solely rely on clinical variables. Notably, hybrid models combining AI predictions with BISAP and mCTSI achieved the highest prognostic performance in our external validation cohort. This synergy suggests a potential staged clinical workflow: (1) initial risk stratification via clinical scoring (such as BISAP) at presentation; (2) if a CT is acquired, the standalone AI model provides automated, opportunistic risk stratification; and (3) when both clinical data and CT are available, the hybrid model offers the most precise risk stratification. We view these findings as a compelling proof-of-concept, warranting future prospective studies to develop unified multimodal risk calculators.

This model is not intended to promote indiscriminate imaging. Adherence to guidelines remains paramount. While early CECT is not recommended by the American College of Gastroenterology for all patients with AP,[30] its use among patients presenting to ED with abdominal pain has increased substantially, reaching 37.8% in 2016[31] and continuing to rise.[32] When admission CT is performed, our model offers a mechanism for opportunistic risk stratification. In this envisioned workflow, the AI model would be applied automatically to CECT scans upon transfer to the PACS, providing a risk triage signal within minutes to assist the treating team.

In our retrospective simulation, the model identified 73% of patients who progressed to SAP as high-risk. However, the PPV of 35.7% indicates that approximately two-thirds of high-risk flags will not develop severe disease. This tradeoff reflects a prioritization of sensitivity for a high-mortality condition; consequently, high-risk outputs should be viewed as alerts that prompt increased vigilance and closer monitoring, rather than triggers for automatic invasive intervention. This framing is essential to mitigate alarm fatigue and prevent resource overuse. Conversely, while the model demonstrated high NPV, the misclassification of a severe case in the external test set highlights that low-risk predictions should not preclude escalation of care if the clinical picture deteriorates.

Previous studies have explored CT-based machine learning approaches for predicting AP severity, using either radiomic features[14–16] or deep neural networks,[12,13,16] with internal AUROCs ranging from 0.77 to 0.92. Salient work includes *PrismSAP*, a multimodal model integrating CECTs, clinical variables, and handcrafted radiomic CT features, which reported an AUROC of 0.916 in an external validation cohort.[16] Although promising, most existing models[14–16] require manual segmentation of the pancreas or slice selection, which limits scalability, introduces reader variability, and hampers real-time deployment. Our study builds on this work by developing a fully automated, end-to-end AI model that operates directly on CECT scans without manual preprocessing, facilitating fast and reliable clinical application.

In addition, this study extends prior research by improving model generalizability. Most existing models were developed and tested at a single institution, limiting evaluation across diverse imaging environments.[12–15] Our ablation study confirms that SSL pretraining on unlabeled data enhances representation transferability, allowing the model to generalize from a U.S. academic health system to a multicenter Hungarian dataset with different acquisition protocols. Notably, while our model leveraged ∼12 000 unlabeled scans, recent literature suggests that SSL pretraining can be effective with as few as 1000 to 5000 images,[33,34] suggesting that this methodology is accessible to institutions with smaller data volumes.

Despite its contribution, our study has several limitations. First, this was a retrospective study and did not include cost-effectiveness or outcome analyses. Prospective studies are needed to determine whether AI-assisted triage can reduce ICU admissions, hospital length of stay, or healthcare costs in real-world settings. Second, our enrichment strategy resulted in a distribution that differs from the general population. Mild AP cases comprised 40% of our internal test set, compared to ∼80% typically reported in the literature.[35,36] The prevalence of SAP (12%) remains consistent with literature estimates of 10%-20%.[37,38] The lower prevalence observed in the external cohort (6.4%) reflects the distinction between our retrospective population and the prospective clinical trial, which excluded high-risk demographics.[21] Third, we did not stratify analyses by pancreatitis etiology, which may influence disease course and imaging manifestations, because etiology was incompletely documented in this retrospective cohort. Future studies in larger, well-phenotyped cohorts should evaluate whether model performance varies by etiology and whether etiology-specific modeling strategies improve generalizability. Finally, our model does not incorporate clinical variables, including many easily obtainable laboratory markers such as BUN level, which could provide additional valuable prognostic information and further improve the performance of image-based models. Misclassified cases suggest that, in some patients, pancreatitis severity may be driven by systemic inflammatory responses or host factors that are not yet apparent on CT.

In conclusion, this multicenter study demonstrated that an AI model trained on admission CECT scans can predict AP severity with accuracy comparable to or exceeding existing prognostic tools, while maintaining robust external performance. These findings suggest that AI-assisted CT analysis may support early, automated risk stratification of AP, warranting further prospective and cost-effectiveness studies to establish its clinical utility.

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Author contributions

Yanqi Xu (Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing—original draft, Writing—review & editing), Brigitta Teutsch (Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Writing—original draft, Writing—review & editing), Weicheng Zeng (Investigation, Software, Validation, Visualization, Writing—original draft, Writing—review & editing), Yang Hu (Conceptualization, Data curation, Investigation, Software, Validation, Writing—original draft, Writing—review & editing), Shikhar Rastogi (Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Writing—review & editing), Emmy Yuebi Hu (Conceptualization, Data curation, Resources, Writing—review & editing), Isabella Mary DeGregorio (Conceptualization, Data curation, Resources, Writing—review & editing), Wan Fung Chui (Conceptualization, Data curation, Resources, Writing—review & editing), Benjamin I. Richter (Conceptualization, Data curation, Resources, Writing—review & editing), Ryan Cummings (Conceptualization, Data curation, Resources, Writing—review & editing), Julia E. Goldberg (Conceptualization, Data curation, Resources, Writing—review & editing), Edwin Mathieu (Conceptualization, Data curation, Resources, Writing—review & editing), Belinda Appiah Asare (Conceptualization, Data curation, Resources, Writing—review & editing), Peter Hegedus (Conceptualization, Investigation, Supervision, Writing—review & editing), Kriszta-Beata Gurza (Conceptualization, Investigation, Writing—review & editing), Istvan Viktor Szabo (Conceptualization, Investigation, Writing—review & editing), Hedvig Tarjan (Conceptualization, Investigation, Writing—review & editing), Andrea Szentesi (Conceptualization, Investigation, Writing—review & editing), Ruben Borbely (Conceptualization, Data curation, Writing—review & editing), Dorottya Molnar (Conceptualization, Data curation, Resources), Nandor Faluhelyi (Conceptualization, Investigation, Writing—review & editing), Aron Vincze (Conceptualization, Investigation, Writing—review & editing), Katalin Marta (Conceptualization, Investigation, Writing—review & editing), Peter Hegyi (Conceptualization, Investigation, Writing—review & editing), Qi Lei (Conceptualization, Investigation, Writing—review & editing), Tamas Gonda (Conceptualization, Data curation, Funding acquisition, Resources, Supervision, Writing—original draft, Writing—review & editing), Chenchan Huang (Conceptualization, Data curation, Funding acquisition, Project administration, Resources, Supervision, Writing—original draft, Writing—review & editing), and Yiqiu Shen (Conceptualization, Data curation, Methodology, Project administration, Resources, Supervision, Writing—original draft, Writing—review & editing)

Supplementary material

Supplementary material is available at *Radiology Advances* online.

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Conflicts of interest

Please see ICMJE form(s) for author conflicts of interest. All authors declare that they have no competing interests or conflicts of interest related to this work.

Data availability

The data underlying this article were obtained from multiple sources. Private imaging and clinical data from NYU Langone Health are not publicly available due to patient privacy and institutional restrictions. Publicly available datasets used in this study—including AMOS, AbdomenCT-1K, the Medical Segmentation Decathlon (MSD), and Synapse—can be accessed from their respective public repositories. Requests for access to data from the GOULASH trial should be directed to the study organizers in accordance with their data access policies.

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